From Causal Discovery to Dynamic Causal Inference in Neural Time Series

Supplementary Materials

Dmitry Zaytsev
Lucy Family Institute for Data & Society
University of Notre Dame
Notre Dame, Indiana, USA
zaytsevdi2@gmail.com

Valentina V. Kuskova
Lucy Family Institute for Data & Society
University of Notre Dame
Notre Dame, Indiana, USA
vkuskova@nd.edu

Michael Coppedge
Political Science
University of Notre Dame
Notre Dame, Indiana, USA
mcoppedg@nd.edu

ACM Reference Format:
Dmitry Zaytsev, Valentina V. Kuskova, and Michael Coppedge. 2026. From Causal Discovery to Dynamic Causal Inference in Neural Time Series: Supplementary Materials. In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2 (KDD '26), August 09–13, 2026, Jeju Island, Republic of Korea.* ACM, New York, NY, USA, 7 pages. https://doi.org/10.1145/3770855.3818956

About this document

This document provides supplementary materials for the paper "From Causal Discovery to Dynamic Causal Inference in Neural Time Series" (KDD '26). It collects the full implementation detail, evaluation-metric definitions, baseline specifications, extended-panel analysis, and the bandwidth-sensitivity, architectural-ablation, and coefficient-path results summarized in the main paper. Section references to "the main paper" refer to the published version. All data and code are available at https://github.com/vkuskova/SIGKDD2026-463.

1 Baseline Model Details

This appendix provides implementation details for all baseline models used in the empirical evaluation. The purpose is transparency and reproducibility. All models are trained and evaluated under the same panel-aware data splits described in Section 5.

1.1 Ridge Vector Autoregression (Ridge VAR)

Model specification. The Ridge VAR baseline is a first-order vector autoregressive model of the form

$$\mathbf{x}_t = \mathbf{B}\,\mathbf{x}_{t-1} + \boldsymbol{\varepsilon}_t, \tag{1}$$

where $\mathbf{x}_t \in \mathbb{R}^N$ denotes the vector of observed variables at time t, $\mathbf{B} \in \mathbb{R}^{N\times N}$ is a constant coefficient matrix, and $\boldsymbol{\varepsilon}_t$ is an innovation term.

The lag order is fixed to one for all experiments.

KDD '26, Jeju Island, Republic of Korea

ACM ISBN 979-8-4007-2259-2/2026/08
https://doi.org/10.1145/3770855.3818956

Estimation and regularization. The coefficient matrix $\mathbf{B}$ is estimated by minimizing the ridge-regularized least squares objective

$$\min_{\mathbf{B}} \sum_{t=2}^{T} \|\mathbf{x}_t - \mathbf{B}\,\mathbf{x}_{t-1}\|_2^2 \;+\; \lambda\,\|\mathbf{B}\|_F^2, \tag{2}$$

where $\lambda > 0$ is a fixed regularization parameter and $\|\cdot\|_F$ denotes the Frobenius norm.

Forecast distribution. Predictive uncertainty is constructed using empirical residual resampling. Let

$$\widehat{\boldsymbol{\varepsilon}}_t = \mathbf{x}_t - \widehat{\mathbf{B}}\,\mathbf{x}_{t-1} \tag{3}$$

denote in-sample residuals. One-step-ahead predictive samples are generated as

$$\widehat{\mathbf{x}}_{t+1}^{(b)} = \widehat{\mathbf{B}}\,\mathbf{x}_t + \widehat{\boldsymbol{\varepsilon}}^{(b)}, \tag{4}$$

where $\widehat{\boldsymbol{\varepsilon}}^{(b)}$ is drawn with replacement from the empirical residual set.

1.2 Time-Varying Vector Autoregression (TV-VAR)

Model specification. The TV-VAR baseline is a locally time-varying first-order VAR of the form

$$\mathbf{x}_t = \mathbf{B}(t)\,\mathbf{x}_{t-1} + \boldsymbol{\varepsilon}_t, \tag{5}$$

where the coefficient matrix $\mathbf{B}(t)$ is allowed to vary smoothly over time.

Kernel-weighted estimation. Local estimates of $\mathbf{B}(t)$ are obtained via kernel-weighted ridge regression. For a target time index t^*, coefficients are estimated by solving

$$\min_{\mathbf{B}} \sum_{s=2}^{T} K\!\left(\frac{s-t^*}{h}\right) \|\mathbf{x}_s - \mathbf{B}\,\mathbf{x}_{s-1}\|_2^2 \;+\; \lambda\,\|\mathbf{B}\|_F^2, \tag{6}$$

where $K(\cdot)$ is a kernel function and $h > 0$ is a bandwidth parameter.

A Gaussian kernel is used:

$$K(u) = \exp\!\left(-\tfrac{1}{2}u^2\right). \tag{7}$$

The bandwidth h is fixed across all experiments.

Estimation window. All observations within the training segment contribute to local estimation, with weights determined by temporal proximity to t^*. No rolling refits or expanding windows are used beyond kernel weighting.

Forecast distribution. Predictive distributions are generated by combining the locally estimated coefficient matrix $\widehat{\mathbf{B}}(t^*)$ with residual resampling, using the same procedure as in Appendix 1.1.

1.3 LSTM with Monte Carlo Dropout

Model specification. The LSTM baseline is a recurrent neural network trained for one-step-ahead prediction. Given an input sequence of length L, the model maps

$$(\mathbf{x}_{t-L}, \ldots, \mathbf{x}_{t-1}) \;\mapsto\; \widehat{\mathbf{x}}_t. \tag{8}$$

The architecture consists of:

- one LSTM layer,
- a fixed hidden state dimension,
- a fully connected linear output layer.

Dropout and training. Dropout is applied to the hidden state with a fixed dropout probability p. The model is trained using mean squared error loss and stochastic gradient descent with adaptive learning rates.

Predictive distribution via Monte Carlo dropout. Predictive uncertainty is approximated using Monte Carlo dropout. At prediction time, dropout remains active and B stochastic forward passes are performed:

$$\widehat{\mathbf{x}}_t^{(b)} = f_{\theta^{(b)}}(\mathbf{x}_{t-L:t-1}), \quad b = 1, \ldots, B, \tag{9}$$

where $\theta^{(b)}$ denotes a stochastic realization of network weights induced by dropout.

The empirical distribution of $\{\widehat{\mathbf{x}}_t^{(b)}\}_{b=1}^{B}$ is used to compute predictive scores and uncertainty measures.

2 Evaluation Metrics and Comparison Measures

This appendix documents the evaluation measures used to compare DCNAR with baseline models in Section 5. The goal of these measures is to assess predictive credibility, uncertainty calibration, and causal behavior under perturbation. All metrics are computed consistently across models using identical training–evaluation splits.

2.1 Multi-horizon predictive distribution accuracy (CRPS)

Predictive distribution accuracy is evaluated using the Continuous Ranked Probability Score (CRPS), a proper scoring rule for probabilistic forecasts. For a scalar outcome y_t with predictive cumulative distribution function F_t, the CRPS is defined as

$$\mathrm{CRPS}(F_t, y_t) = \int_{-\infty}^{\infty} \left(F_t(z) - \mathbb{I}\{z \geq y_t\}\right)^2 \, dz. \tag{10}$$

Equivalently, when F_t is represented by an empirical predictive distribution with samples $\{X_t^{(b)}\}_{b=1}^{B}$, CRPS can be written as

$$\mathrm{CRPS}(F_t, y_t) = \mathbb{E}\left|X_t - y_t\right| - \frac{1}{2}\mathbb{E}\left|X_t - X_t'\right|, \tag{11}$$

where X_t and X_t' are independent draws from F_t.

For multi-horizon evaluation, predictive distributions are generated at horizons $h = 1, \ldots, H$, and CRPS values are averaged across horizons, variables, countries, and forecast origins.

2.2 Local one-step distributional accuracy

To assess short-horizon predictive behavior independently of long-run dynamics, we also compute local one-step-ahead CRPS. For each forecast origin t, a one-step predictive distribution F_{t+1} is generated using information available up to time t. The resulting CRPS values are averaged across countries, variables, and forecast origins:

$$\mathrm{CRPS}_{\mathrm{local}} = \frac{1}{NCT} \sum_{c=1}^{C} \sum_{i=1}^{N} \sum_{t \in \mathcal{T}} \mathrm{CRPS}\left(F_{t+1}^{(c,i)}, x_{t+1}^{(c,i)}\right), \tag{12}$$

where C denotes the number of countries, N the number of variables, and $\mathcal{T}$ the set of evaluation times.

This metric isolates local distributional accuracy without conflating it with longer-horizon stability or propagation effects.

2.3 Prediction interval coverage

Uncertainty calibration is assessed via empirical coverage of nominal prediction intervals. For each predictive distribution F_t, we construct a $(1 - \alpha)$ prediction interval

$$\left[Q_{\alpha/2}(F_t),\ Q_{1-\alpha/2}(F_t)\right],$$

where $Q_q(F_t)$ denotes the q-th quantile of the predictive distribution.

Empirical coverage at horizon h is defined as

$$\mathrm{Coverage}(h) = \frac{1}{NC} \sum_{c=1}^{C} \sum_{i=1}^{N} \mathbb{I}\left\{x_{t+h}^{(c,i)} \in \left[Q_{\alpha/2}(F_{t+h}^{(c,i)}),\ Q_{1-\alpha/2}(F_{t+h}^{(c,i)})\right]\right\}, \tag{13}$$

averaged across forecast origins t.

We report coverage for $\alpha = 0.10$, corresponding to nominal 90% prediction intervals.

2.4 Representative counterfactual impulse responses

Causal behavior is evaluated using impulse responses and counterfactual trajectories derived from each fitted model. For a given forecast origin t_0 and shock variable j, we construct a counterfactual trajectory by applying an intervention $\boldsymbol{\delta}_t$ to the system dynamics.

Let $\mathbf{x}_t^{(0)}$ denote the baseline trajectory generated by the fitted model, and $\mathbf{x}_t^{(\delta)}$ the trajectory under intervention. A one-time unit shock at horizon $h = 1$ is defined as

$$\boldsymbol{\delta}_t = \begin{cases} \delta\, \mathbf{e}_j, & t = t_0 + 1, \\ \mathbf{0}, & \text{otherwise}, \end{cases}$$

where $\mathbf{e}_j$ is the j-th canonical basis vector and δ is scaled to the empirical standard deviation of variable j.

The counterfactual impulse response for variable i at horizon h is then defined as

$$\mathrm{IRF}_{i \leftarrow j}(t_0, h) = x_{i,t_0+h}^{(\delta)} - x_{i,t_0+h}^{(0)}. \tag{14}$$

For system-level analysis, we summarize counterfactual deviation using the L^2 norm across variables:

$$\mathcal{R}(t_0 + h) = \left\| \mathbf{x}_{t_0+h}^{(\delta)} - \mathbf{x}_{t_0+h}^{(0)} \right\|_2. \tag{15}$$

Representative impulse responses and system-level counterfactual trajectories are selected for illustrative comparison across models in Section 5.

3 Implementation Details for DCNAR

This appendix documents implementation details of DCNAR to support reproducibility. All design choices reported here correspond to the experiments described in Section 5.

3.1 Causal Discovery Stage

NAVAR variant. The causal discovery stage is instantiated using a neural additive vector autoregression (NAVAR) model with additive, variable-specific neural components. Each target variable is modeled independently using a feedforward neural network with convolutional preprocessing over lagged inputs (MLP/Conv variant). No recurrent components are used.

Lag length. The maximum lag length is fixed to

$$L = 8, \tag{16}$$

and identical lag structure is used for all variables and all countries.

Regularization. Sparsity in the learned causal graph is encouraged through ℓ_1 regularization applied to the additive component functions. Specifically, regularization is applied to the contribution magnitudes of the neural functions $f_{ij\ell}$, encouraging many directed interactions to shrink toward zero. Weight decay is additionally applied to stabilize neural optimization.

Normalization. No normalization is applied to the input time series prior to NAVAR training. Likewise, no normalization is applied to the learned causal score matrix. All causal scores are therefore expressed in the native scale of the data and are treated as relative, not absolute, measures of influence.

Output. The output of the causal discovery stage is a directed causal score matrix

$$\mathrm{S} \in \mathbb{R}^{N \times N},$$

aggregated across lags. This matrix is treated as a structural hypothesis rather than as an inferential object. Additional processing of S (e.g., necessity filtering, stability screening) is described below.

3.2 Construction of the Structural Prior via Edge Ablation

The purpose of refining the causal adjacency matrix is to obtain a sparse structural prior that prioritizes directed relationships which could materially affect model behavior. A detailed methodological treatment and evaluation of this approach is provided in [3]; here we document its role in the present experiments.

Edge ablation procedure. Starting from an initial directed adjacency matrix inferred during the causal discovery stage, we consider the effect of selectively removing individual directed edges. For each ordered pair (j, i) such that the initial adjacency matrix indicates a potential connection, we construct an ablated variant of the network in which that single edge is removed while all other edges are retained.

For each ablated network, the dynamic model is re-estimated using the same specification and hyperparameters as the full model, differing only in the exclusion of the selected edge. This yields a family of models that are identical except for the presence or absence of a single directed interaction.

Forecast-based comparison. To assess the impact of edge removal, we compare out-of-sample predictive behavior of the full model and each ablated variant using a fixed forecasting protocol. Let $e_t^{(0)}$ denote the forecast error at time t under the full model and $e_t^{(-ij)}$ the corresponding error under the model in which edge (j, i) has been removed. The effect of ablation is summarized by the loss differential

$$d_t^{(ij)} = L(e_t^{(-ij)}) - L(e_t^{(0)}),$$

where $L(\cdot)$ is a fixed loss function.

Positive average loss differentials indicate that removal of the edge degrades predictive performance, while negligible or negative differentials suggest that the edge does not materially affect model behavior.

Statistical assessment. Statistical significance of loss differentials is evaluated using Diebold–Mariano tests [1], which account explicitly for serial dependence in forecast errors. These tests assess whether the predictive performance of the ablated model differs systematically from that of the full model over the evaluation period.

Edges whose removal leads to statistically significant degradation in predictive performance are retained in the structural prior. Edges whose removal has little or no effect are excluded. The resulting adjacency matrix is therefore weighted and sparse, reflecting predictive necessity rather than coefficient magnitude.

Dynamic coherence screening. As a supplementary diagnostic, we examine whether the resulting structural prior yields stable and interpretable time-varying dynamics. Specifically, we assess whether estimated causal influence trajectories exhibit smooth temporal evolution rather than erratic fluctuation. Structures that produce highly unstable or noisy dynamics are treated as unreliable and excluded from further analysis.

Role in the present paper. In the experiments reported in this paper, the structural prior supplied to the dynamic inference stage is the weighted adjacency matrix obtained via this edge ablation procedure. The procedure is used solely to select a plausible and empirically grounded structural prior; the substantive evaluation of edge necessity, stability, and coherence is addressed in detail in [3].

3.3 Dynamic Inference Stage

tvNAR formulation. Dynamic causal inference is performed using a time-varying network autoregressive model [2] conditioned on a learned structural prior. The primary specification used in the main text is tvNAR(1), which takes the form

$$\mathbf{x}_t = (\widehat{\mathbf{G}} + \mathbf{I})\, \mathbf{\Lambda}(t)\, \mathbf{x}_{t-1} + \boldsymbol{\varepsilon}_t,$$

where $\widehat{\mathbf{G}}$ is the learned adjacency matrix and $\mathbf{\Lambda}(t)$ is a diagonal matrix of time-varying node influence parameters.

tvNAR(p). The implementation supports higher-order dynamics via tvNAR(p),

$$\mathbf{x}_t = \sum_{\ell=1}^{p} (\widehat{\mathbf{G}} + \mathbf{I})\, \mathbf{\Lambda}_\ell(t)\, \mathbf{x}_{t-\ell} + \boldsymbol{\varepsilon}_t,$$

of which tvNAR(1) is a special case. In practice, tvNAR(1) is used in the main experiments, while tvNAR(p) is explored in supplementary analyses.

Kernel smoothing. Time variation in $\mathbf{\Lambda}(t)$ is estimated using kernel-weighted local regression over a normalized time index $\tau \in (0, 1)$. A Gaussian kernel is used:

$$K(u) = \exp\left(-\tfrac{1}{2}u^2\right),$$

with a fixed bandwidth parameter. Kernel weights are computed within each country series and then pooled across countries using panel-aware indexing.

Time indexing in panel data. Each country time series is mapped to a normalized time index

$$\tau_{c,t} = \frac{t}{T_c},$$

where T_c is the length of the series for country c. Kernel smoothing is performed with respect to $\tau_{c,t}$, allowing time variation to be estimated consistently across panels of equal length. All panel splitting and indexing respect country boundaries; no temporal leakage occurs across units.

3.4 Computational Considerations

Runtime. The computational cost of DCNAR is dominated by the causal discovery stage. NAVAR training scales approximately as

$$O(N^2 \cdot L \cdot T),$$

where N is the number of variables, L the lag length, and T the total number of time points across panels. In the experiments reported here, NAVAR training completes within two-three minutes on a single GPU. The dynamic inference stage scales linearly in T for fixed N and is computationally lightweight relative to NAVAR.

Memory usage. Memory requirements are modest. NAVAR requires storing lagged input tensors and per-variable neural models, while tvNAR requires only the learned adjacency matrix and time-varying coefficient paths. All experiments fit comfortably within standard GPU memory constraints.

Parallelization. The implementation supports parallelization across countries at both stages. NAVAR training is parallelized implicitly through batched optimization, while tvNAR estimation and counterfactual simulation are embarrassingly parallel across panels and forecast horizons. All reported experiments were run using standard multi-core CPU resources and a single GPU.

Reproducibility. All hyperparameters, data splits, and model configurations are fixed across experiments. No manual tuning is performed per country or per variable. The full experimental pipeline is deterministic up to stochastic neural optimization and Monte Carlo sampling used for uncertainty estimation.

4 Extended Panel Analysis Setup

This appendix documents the data construction and experimental protocol for the extended panel analysis used to assess the robustness of DCNAR to increased temporal support. Results corresponding to this setup are reported in Appendix 4.4.

4.1 Extended Dataset Construction

The extended panel is derived from the Varieties of Democracy (V-Dem) country–year dataset and uses the same set of democracy components as the main analysis. Variable definitions, coding procedures, and substantive interpretation are identical to those described in Section 5.

Unlike the main panel, which prioritizes breadth across countries, the extended panel prioritizes temporal depth. Countries are retained only if they exhibit complete, uninterrupted coverage across all selected variables for a substantially longer time span. This requirement leads to a smaller but temporally richer sample. The resulting dataset consists of 89 countries, observed annually for $T = 75$ consecutive years, with no missing values across the selected democracy components.

This construction yields a strongly balanced panel with approximately twice the temporal length of the main analysis but fewer cross-sectional units. No additional filtering or transformation is applied beyond the criteria above.

4.2 Motivation for Extended Panel Analysis

The extended panel serves as a robustness check for dynamic causal inference under substantially different data conditions. While the main analysis reflects the standard setting in empirical democracy research, where data are typically represented by many countries with relatively short time series, the extended panel approximates a complementary regime with longer temporal trajectories but reduced cross-sectional diversity.

Evaluating DCNAR under both configurations allows us to assess whether its inferred dynamic causal behavior depends critically on the short-panel setting or whether it reflects more persistent features of the data-generating process. In particular, the extended panel tests whether impulse responses and counterfactual trajectories remain stable when substantially more temporal information is available for each unit.

4.3 Experimental Protocol

The experimental protocol for the extended panel mirrors that of the main analysis as closely as possible. All modeling choices, hyperparameters, and evaluation procedures are held fixed.

Specifically:

- The same causal discovery procedure is applied to the extended panel without retuning.
- The same DCNAR dynamic inference configuration is used.
- The same baseline models (Ridge VAR, TV-VAR, and LSTM with Monte Carlo dropout) are evaluated.
- Training–evaluation splits are defined analogously, with each country series divided into a training segment and a held-out evaluation segment.

No model parameters are adjusted to account for the increased temporal length. This ensures that differences in behavior can be attributed to data characteristics rather than to changes in model specification.

4.4 Results

This section reports results for the extended panel analysis based on the 75-year sample and interprets them in relation to the main results presented in Section 5 of the main paper. The purpose is to assess the robustness of DCNAR's dynamic causal behavior under substantially increased temporal support.

Predictive and distributional performance. Panels (A)–(C) of Figure 1 summarize predictive distribution accuracy and calibration across models. As in the main analysis, DCNAR achieves predictive performance comparable to linear and time-varying VAR baselines across forecast horizons. Differences in CRPS and one-step distributional accuracy are modest, and no model consistently dominates across all horizons. Empirical coverage of nominal 90% prediction intervals remains stable for DCNAR across horizons, while the LSTM baseline again exhibits undercoverage.

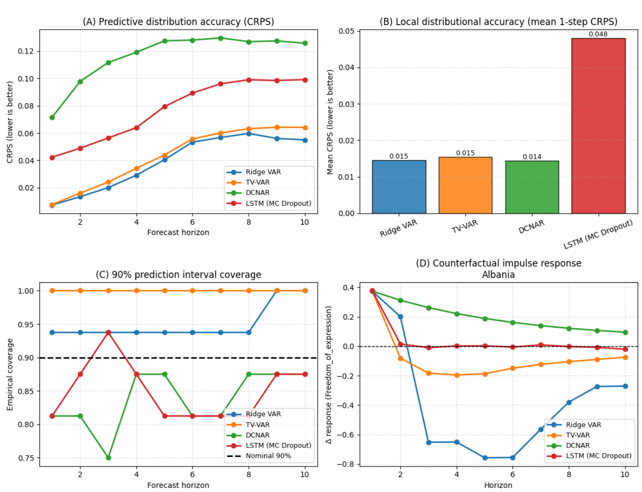

Figure 1: Comparison of DCNAR on the 89-country, 75-year panel, with Ridge VAR, TV-VAR, and LSTM (MC Dropout) across predictive (panel A, CRPS), distributional (panel B, local distributional accuracy - mean one-step-ahead CRPS), and causal diagnostics (panel C, nominal 90% prediction intervals across horizons) on the extended-panel democracy dataset. Panel D shows representative counterfactual impulse response following a positive shock to freedom of expression (Albania).

Impulse response and counterfactual behavior. Panel (D) of Figure 1 presents a representative counterfactual impulse response for Albania following a positive shock to freedom of expression; the response is smooth, monotonic, and gradually decaying, while Ridge VAR and TV-VAR again exhibit oscillations and sign reversals. Figure 2 extends the analysis to system-level counterfactual trajectories, summarized using the L^2 norm across all variables. Under DCNAR, counterfactual paths diverge smoothly from baseline and remain bounded over the forecast horizon, with cross-country magnitude differences reflecting known institutional heterogeneity.

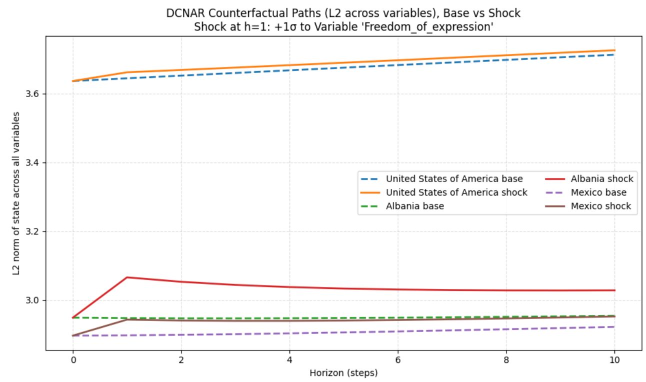

Figure 2: System-level counterfactual trajectories under DCNAR on the 89-country, 75-year panel, summarized by the L^2 normalization across all democracy components, for Albania, the United States, and Mexico. Solid lines denote trajectories under a localized positive shock to freedom of expression at horizon $h = 1$; dashed lines denote corresponding baseline trajectories without intervention.

The persistence of these qualitative differences in a substantially longer panel indicates that DCNAR's behavior is not an artifact of short time series or limited temporal resolution.

Implications for stability. Taken together, the extended panel results reinforce the central claim of the paper: DCNAR yields stable and interpretable dynamic causal behavior across data regimes with very different temporal characteristics. The consistency of impulse responses and counterfactual trajectories across the 35-year and 75-year panels suggests that DCNAR captures persistent features of the underlying data-generating process rather than overfitting to a particular sample window.

These findings support the interpretation of DCNAR as a robust methodological instrument for dynamic causal analysis in observational panel settings, rather than as a model whose conclusions are sensitive to specific data configurations.

4.5 Scope and Interpretation

The purpose of the extended panel analysis is not to improve predictive performance or to optimize model fit under favorable conditions. Rather, it is intended to assess the stability and robustness of dynamic causal behavior, in particular, impulse responses and counterfactual trajectories, when the amount of temporal information available per unit is substantially increased.

Results presented in Appendix 4.4 should therefore be interpreted qualitatively, in comparison to the main analysis, with emphasis on consistency of dynamic patterns rather than on absolute performance metrics.

5 Bandwidth Sensitivity Analysis

To verify that the smoothness of DCNAR's impulse responses reflects genuine temporal regularization rather than an artifact of the kernel bandwidth, we repeated the analysis across six bandwidth values spanning highly local to near-global smoothing. The kernel is one-sided throughout: the estimate at time t uses only observations at times $s \leq t$, so no future information contributes to any coefficient estimate. Table 1 reports mean CRPS and impulse-response diagnostics; Figure 3 visualizes the CRPS values.

Table 1: Bandwidth sensitivity. Mean CRPS varies by less than 2×10^{-4} across the full range, and impulse responses remain smooth, monotonic, and sign-consistent (zero sign changes) for every country and bandwidth.

Bandwidth h	Mean CRPS	Sign consistent	Monotonic	Sign changes
0.05	0.01616	Yes	Yes	0
0.10	0.01622	Yes	Yes	0
0.15	0.01624	Yes	Yes	0
0.20	0.01626	Yes	Yes	0
0.30	0.01629	Yes	Yes	0
0.40	0.01630	Yes	Yes	0

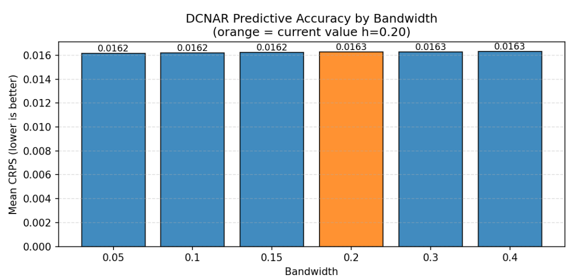

Figure 3: DCNAR predictive accuracy (mean CRPS) across bandwidth values. The bandwidth used in the main experiments ($h = 0.20$) is highlighted. Predictive accuracy is essentially flat across the full range.

The impulse-response diagnostics are identical across all bandwidths and all three countries examined (Albania, the United States, and Mexico): smooth, monotonic decay with a peak at horizon $h = 1$ and zero sign changes. This confirms that the qualitative dynamics reported in the main text are a feature of the estimated system rather than a consequence of the smoothing choice.

6 Architectural Ablation

This section provides the full architectural ablation summarized in Table 1 (Section 5) of the main paper. Each DCNAR component is removed in turn while all others are held fixed: (i) the neural causal-discovery module is replaced by a Ridge VAR–derived adjacency at matched density; (ii) the time-varying $\Lambda(t)$ is replaced by its static, time-averaged counterpart; and (iii) necessity-based sparsity is removed, yielding a dense $\widehat{G}$. Table 2 reports predictive accuracy and qualitative impulse-response diagnostics; Figure 4 shows the corresponding per-country impulse responses.

Table 2: Architectural ablation on the short panel. Replacing neural discovery with a linear VAR–derived structure degrades CRPS by 14.5%. All variants preserve sign-consistent, monotonic impulse responses across the three countries examined.

Condition	Edges	Λ	CRPS	Sign / Mono.
DCNAR (full pipeline)	166	time-varying	0.01311	3/3 / 3/3
VAR-derived $\widehat{G}$	166	time-varying	0.01502	3/3 / 3/3
Static Λ	166	static	0.01306	3/3 / 3/3
Dense $\widehat{G}$	240	time-varying	0.01323	3/3 / 3/3

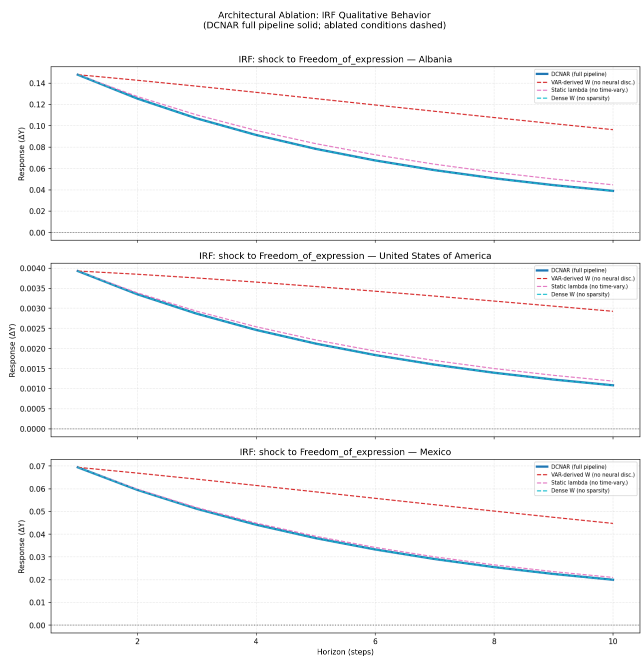

Figure 4: Impulse-response behavior under each ablation condition (DCNAR full pipeline solid; ablated conditions dashed) for shocks to freedom of expression in Albania, the United States, and Mexico. The VAR-derived structure systematically overestimates shock persistence (slower decay), while the other variants closely track the full pipeline.

The neural discovery stage is the component with the clearest predictive contribution: substituting a linear VAR–derived structure degrades CRPS by 14.5% and produces impulse responses that systematically overestimate shock persistence. Removing time variation leaves one-step CRPS essentially unchanged (indeed marginally lower at 0.01306), which is expected in a highly persistent panel where short-horizon prediction is insensitive to smooth coefficient drift; the contribution of $\Lambda(t)$ is visible instead in longer-horizon causal behavior. Removing sparsity modestly worsens CRPS, consistent with necessity-based filtering removing redundant propagation paths. Because $\widehat{G}$ and $\Lambda(t)$ enter jointly through their product, these results should be read as sensitivity analyses rather than fully identified component effects.

7 Time-Varying Coefficient Paths and Effective Networks

This section provides the coefficient-path and effective-network visualizations referenced in Section 6 of the main paper. Without any domain supervision, the estimated node-influence paths $\lambda_i(t)$ recover historically documented regime dynamics, and they do so more coherently than the corresponding TV-VAR diagonal coefficients.

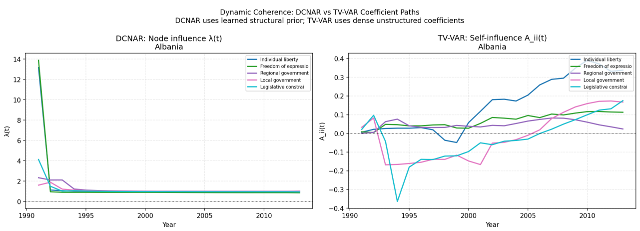

Figure 5: Dynamic coherence: DCNAR node-influence paths $\lambda(t)$ (left) versus TV-VAR self-influence coefficients $A_{ii}(t)$ (right) for Albania. DCNAR uses the learned structural prior; TV-VAR uses dense, unstructured coefficients. DCNAR paths stabilize after the early-1990s transition, whereas TV-VAR coefficients remain comparatively erratic.

For Albania, DCNAR influence parameters fall sharply over 1991–1993 (post-communist institutional collapse) before stabilizing near unity; for the United States they remain near-constant (institutional stability); and for Mexico they show a modest shift around 2000, coinciding with the end of single-party rule. By contrast, the TV-VAR diagonal coefficient paths reach implausibly large magnitudes for Mexico and Albania that are difficult to reconcile with domain knowledge.

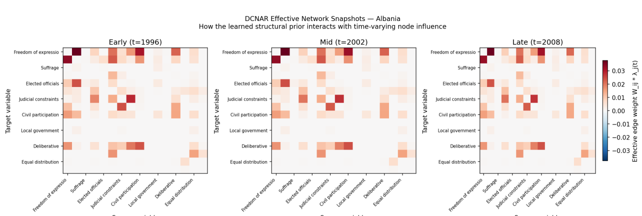

Figure 6: DCNAR effective network snapshots for Albania at three time points (1996, 2002, 2008), showing the effective edge weights $W_{ij}\,\lambda_j(t)$. The learned structure remains stable across the observation window while node influence evolves smoothly.

Effective-network snapshots (Figure 6) confirm that the learned structure remains stable across 1996, 2002, and 2008 while node influence evolves smoothly. This kind of substantive, historically interpretable read-out is inaccessible to static causal-discovery methods or black-box predictive models.

References

[1] F. X. Diebold. 2016. Comparing predictive accuracy, twenty years later: A personal perspective on the use and abuse of Diebold–Mariano tests. *Journal of Business & Economic Statistics* 33, 1 (2016), 1.

[2] Y. Ding, X. Zhu, R. Pan, and B. Zhang. 2025. Network Vector Autoregression with Time-Varying Nodal Influence. *Computational Economics* (2025), 1–27.

[3] V. Kuskova, D. Zaytsev, and M. Coppedge. 2026. Beyond Coefficients: Forecast-Necessity Testing for Interpretable Causal Discovery in Nonlinear Time-Series Models. In *Proceedings of the 39th International Florida Artificial Intelligence Research Society Conference (FLAIRS).* doi:10.32473/flairs.39.1.141791 arXiv:2604.18751.